KINETIC THEORY AND HEAT

The Zeroth Law of Thermodynamics

The **zeroth law of thermodynamics** states that if two systems are both in thermal equilibrium with a third system, then the first two systems are also in thermal equilibrium with each other.

In simple terms the Zeroth Law of Thermodynamics states that systems that are in thermal equilibrium exist at the same temperature. Figure 1 shows three systems in thermal equilibrium. Let A, B, and C be three systems. If A and C are in thermal equilibrium, and A and B are in thermal equilibrium, then B and C are in thermal equilibrium.

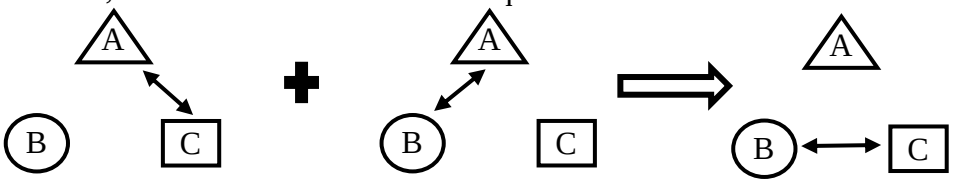


Figure 1. The double arrow represents thermal equilibrium between systems. If systems A and C are in equilibrium, and systems A and B are in equilibrium, then systems B and C are in equilibrium. The systems A, B, and C are at the same temperature.

Systems are in thermal equilibrium if they do not transfer heat, even though they are in a position to do so, based on other factors. For example, food that is stored in a refrigerator overnight is in thermal equilibrium with the air in the refrigerator: heat no longer flows from one source (the food) to the other source (the air) or back.

Temperature and Temperature Scales

The temperature of a substance is a number which expresses its degree of hotness on some chosen scale. Thus, the temperature of water fetched from the tap is lower than that of a boiling water; the boiling water contains a much larger quantity of internal energy than the tap water. Temperature is measured by means of a thermometer, of which a number of different types are available, but each makes use of a particular thermometric property of a particular thermometric substance. Some depend on the uniform expansion of a liquid when heated; others on the linear uniform expansion of a compound strip of two metals. In order to establish a temperature scale it is necessary to make use of fixed points: A fixed point is the single temperature at which it can confidently be expected that a particular physical event (e.g. the melting of ice under specific conditions) always takes place. Three such points are defined below.

1. The upper fixed point or steam point is the temperature of steam from water boiling under standard atmospheric pressure of $760\,mmHg$. It is labelled 100 ^{o}C or 373 K.
2. The lower fixed point or ice point is the temperature of pure melting ice. It is labelled 0 ^{o}C or 273 K.
3. The triple point of water is that unique temperature at which pure ice, pure water and water vapour can exist together in equilibrium. The triple point of water is 273.16 K, or 0.01 °C, or 32.018 °F.

The triple point is particularly useful, since there is only one pressure at which all three phases (solid, liquid and gas) can be in equilibrium with each other.
The difference in temperature between the two fixed points is called the *fundamental interval* as shown in Figure 2. This is divided into 100 equal degrees, the ice point being called 0 ^{0}C and the steam point 100 ^{0}C. This subdivision of the interval was suggested by a Swedish astronomer named Celsius and is now called the *Celsius scale*.

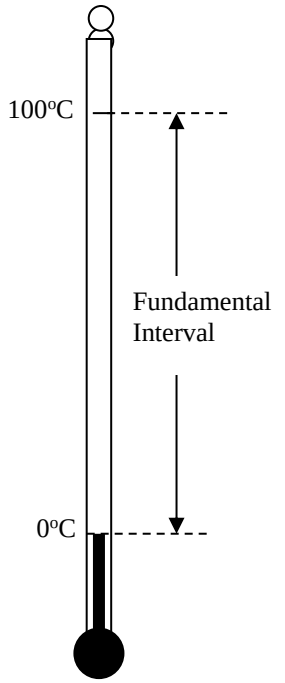


Figure 2. A thermometer showing the upper and lower points.

The SI unit of temperature is the Kelvin (K). An interval of one Kelvin is defined as being $\frac{1}{273.16}$ of the temperature of the triple point of water as measured on the thermodynamic scale of temperature.
The triple point of water is the fixed point of the scale and is assigned the value of 273.16 K. On this basis absolute zero is 0 K, the ice point is 273.15 K, and the steam point is 373.15 K.

Equation 4.28 gives the relationship between the Celsius scale and the Kelvin scale

$$\theta = T - 273.15 \qquad 1$$

where, θ = temperature in ^{0}C, and T = temperature in K.

It follows from equation 7.1 that a temperature change of 1K is exactly equal to a temperature change of 1 ^{0}C.

THERMOMETERS, PROPERTIES OF THERMOMETRIC MATERIALS AND TYPES OF THERMOMETERS

In this session you are introduced to the thermometer as a device that measures temperature or a temperature gradient. The thermometer has two important elements which makes it possible to read temperature of surroundings. The properties of a good thermometric material have been outlined. Reasons for chosing mercury instead of water as a thermometric liquid is also outlined. Calibration of liquid-in-glass thermometers have also been discussed. Some commonly used types of thermometers are discussed.

Objectives

At the end of this session you should be able to:

1. Identify a thermometer and describe its usage.
2. The properties of a good thermometric material.
3. Explain why some materials cannot be used as thermometric materials.
4. Calibrate a thermometer.
5. Name some existing thermometers.
6. Describe some types of thermometers.
7. State the properties of alcohol and mercury which qualifies them as good thermometric liquids.

Thermometers

A thermometer is a device that measures temperature or a temperature gradient. A thermometer has two important elements:

1. A temperature sensor. In liquid-in-glass thermometers, the bulb of thermometric liquid is the sensor in which a change occurs when there is change in temperature.
2. A means of converting this change into numerical value on a visible scale.

Thermometric Materials

Certain substances have the property that changes with temperature. Substances that show a uniform change in temperature can be used as a thermometric material. For example, some substances expand on heating and contract on cooling, some change their colours and some also change their electric resistance. Common liquid-in-glass thermometers are generally made using some suitable liquid like mercury or alcohol as thermometric material.

A thermometric liquid should have the following properties:

i. It should be visible.
ii. It should have uniform thermal expansion.
iii. It should have a low freezing point.
iv. It should have a high boiling point.
v. It should not wet glass.
vi. It should be a good conductor of heat.
vii. It should have a small specific heat capacity
viii. It should respond to temperature change rapidly

Example 1. Give 4 reasons why water is not used as a thermometric liquid.

Solution

1. Water is transparent.
2. Water wets glass
3. Water has high specific heat capacity
4. Water has anomalous expansion

Calibration of Thermometers

Thermometers can be calibrated either by comparing them with other calibrated thermometers or by checking them against known fixed points on the temperature scale. The best known of these fixed points are the melting and boiling points of pure water. (Note that the boiling point of water varies with pressure, so this must be controlled.)

The traditional way of putting a scale on a liquid-in-glass or liquid-in-metal thermometer was in three stages:

1. Immerse the sensing portion in a stirred mixture of pure ice and water at atmospheric pressure and mark the point indicated when it had come to thermal equilibrium.
2. Immerse the sensing portion in a steam bath at Standard atmospheric pressure and again mark the point indicated.
3. Divide the distance between these marks into equal portions according to the temperature scale being used.

Why is mercury used in thermometers instead of water?

Mercury is used mostly in thermometers because of the following properties:

- It is visible.
- It has a low freezing point (-39 °C).
- It has a very high boiling point (357 °C).
- It expands linearly.
- It gives accurate measurements.
- It is a good conductor of heat.
- Fast response time.
- It has a wide range of temperatures.

Properties of Alcohols as a Thermometric Liquid

- It has a low freezing point (-112 °C).
- Alcohol has a low boiling point of about 78ºC and therefore cannot be used to measure high temperatures.
- Alcohol is fairly inexpensive.
- It can be colored.
- It expands more than mercury.

Types of Thermometers

A liquid in glass thermometer has a bulb with a long capillary tube of uniform and fine bore. A suitable liquid is filled in the bulb. When the bulb contacts a hot object, the liquid in it expands and rises in the tube. The glass stem of a thermometer is thick and acts as a cylinder lens. This makes it easy to see the liquid level in the glass tube. Mostly **Mercury** and **alcohol** are used in glass thermometers. There are many types of thermometers, but some common thermometers are laboratory thermometer, clinical thermometer, thermocouple, Bimetallic strip thermometer and Pyrometers.

Laboratory Thermometer

It is used to measure the temperature of solids, liquids and gases in experiments, as well as the temperature of the environment. It measures temperature in the range of -5 °C to 110 °C.

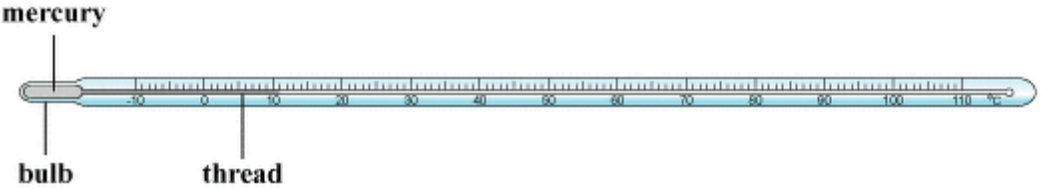


Figure 3 This is a Laboratory Thermometer

Clinical Thermometer

It is used to measure human body temperatures.





Figure 4 This shows different types of clinical thermometer. The second picture uses infrared sensor.

Structure:

- It is smaller in size than the laboratory thermometer.
- It has a constriction in the fine capillary tube. This constriction prevents the mercury from contracting and flowing back when it is removed from human body. This enables the reading and recording of the maximum temperature of the body without haste.
- The mercury thread needs to be shaken back into the bulb before the thermometer is used again.

Range: The scale ranges between 35°C to 42°C.
This range is centered around the normal body temperature of 37°C. Short range enables the scale to be divided into smaller intervals for greater accuracy (0.1°C).
Digital thermometers are now available in the hospitals. These are advanced thermometers used to measure the temperature of a body with a high level of accuracy.

HEAT ENERGY, HEAT CAPACITY, SPECIFIC HEAT CAPACITY AND LATENT HEAT

In this session internal energy and heat energy are explained. heat capacity and specific heat capacity are also explained and well defined. Formulas for calculating heat energy, heat capacity and specific heat capacity are stated. Also, latent energy is explained. Latent heat of fusion and latent heat of vaporization are also explained and well defined. Formulas for calculating latent heat energy are stated.

Objectives

At the end of this session you should be able to:

1. define heat energy and state its units.
2. explain and define heat capacity of a substance.
3. explain and define specific heat capacity of a substance.
4. calculate the amount of heat in a body at a given temperature.
5. explain latent heat energy and state its units.
6. explain and define latent heat of fusion of a substance.
7. explain and define latent heat of vaporization of a substance.
8. solve problems involving latent heat and sensible heat.

Heat Energy

Heat is a measure of the total kinetic energy of the atoms or molecules in a body. Heat is a form of energy measured in Joules (J). The heat content of a body will depend on its temperature, its mass, and the material it is made of. Heat energy is always transferred from an object at high temperature to one at lower temperature.

Internal energy is all of the energy belonging to a system while it is stationary. Thermal energy is the portion of internal energy that changes when the temperature of a system changes.
In physics, heat is defined as the energy which is transferred from one place to another owing to a temperature difference between them. When you heat a substance, you are transferring energy into

it by placing it in contact with surroundings that have a higher temperature. A typical example is when a sauce pan containing cold water is placed on a stove burner. The burner will transfer heat to the sauce pan and the cold water, and after some time, the sauce pan and cold water will become very hot.

Heat Capacity and Specific Heat Capacity

When heat energy is transferred to a body, its temperature rises but the state remains unchanged. The quantity of heat energy required to raise the temperature of a given mass of a substance by one degree Celsius varies from one substance to another.

For any mass of a particular substance, the quantity of heat required to raise its temperature by 1^oC is referred to as the *Heat Capacity* of that substance. For example, if the initial temperature of a metal rod is 26 ^{0}C, then the heat energy required to raise its temperature to 27 ^{0}C is its Heat Capacity. Heat capacity is denoted by *C* and its SI unit is J/K.

Heat Capacity, $Q = C\left(T_f - T_i\right) = C\Delta T$ **2**

Or

$Q = C\left(\theta_f - \theta_i\right) = C\Delta\theta$

where θ_f and θ_i are the final and initial temperatures of the object in degrees Celsius and also T_f and T_i are the final and initial temperatures of the object in Kelvin.

Specific Heat Capacity

We know that when heat energy is absorbed by a substance, its temperature increases. If the same quantity of heat is given to equal masses of different substances, it is observed that the rise in temperature for each substance is different. This is due to the fact that different substances have different heat capacities. For example, the quantity of energy needed to raise the temperature of one 1 kg of water by 1 ^{0}C is 4186 J, but the quantity of energy needed to raise the temperature of 1 kg of copper by 1 ^{0}C is only 387 J.

Specific heat capacity (*c*) is defined as the quantity of heat energy required to raise the temperature of 1 Kg of a substance by 1^0 C.

Mathematically, quantity of heat can be expressed as;

$Q = c\ m\ \Delta T$ **3**

where *Q* is the quantity of heat absorbed by a body, *m* is its mass, ΔT is the change in temperature and *c* is the specific heat capacity.

Note: The specific heat capacity of a substance depends on the nature of the material the substance is made from. S.I unit of specific heat J $kg^{-1}K^{-1}$

Note: Heat capacity = specific heat x mass

That is, $C = mc$.

Example 2. What is the heat energy required to raise the temperature of 0.5 kg of water by 3 ^{0}C?

Solution

$Q = mc\Delta\theta = (0.5kg)(4186\,J/kg.^oC)(3^oC) = 6.28\times10^3\,J$.

Example 3. Find the temperature change if 2500J of heat is supplied to 200 g of water. [take SHC of water to be 4186 J/kg/K].

Solution

$$\Delta\theta = \frac{Q}{mc} = \frac{2500}{0.2 \times 4186} = 2.99K$$

Example 4. A 300 g of cold water at 20° C was mixed with a 400 g of hot water at a temperature of 65° C. Find the final temperature of the mixture.

Solution

Heat lost by hot water = Heat gained by cold water

$Q_h = Q_c$

$0.4\times4186\times(65-\theta) = 0.3\times4186\times(\theta-20)$

$26-0.4\theta = 0.3\theta-6$

$26+6 = (0.4+0.3)\theta$

$\theta = \frac{32}{0.7} = 45.7\ ^o\text{C}$

Note: When the temperature increases, Q and ΔT are both taken to be positive and energy flows into the system. When the temperature decreases, Q and ΔT are both negative, and energy flows out of the system.

Latent Heat

Another important effect of heat is that it can produce a change in a body's state. As we know matter can exist in any of the three states- solid, liquid or gas. A change of matter from one physical state to another is called change of state. The state of matter depends on the temperature and the pressure that is exerted on it. The quantity of heat energy required to change the temperature of matter but not necessarily changing its state is called *Sensible Heat.* There are other cases where heat energy absorbed by or removed from matter does not change its temperature but changes its state. Such energy is referred to as hidden heat or latent heat. This energy is called hidden or latent because instead of the heat raising the temperature of the matter, it rather works on the molecules breaking them further apart. *Latent heat* is the quantity of heat required to change completely the state of matter to another state without any change in temperature. A typical example is the quantity of heat required to completely change 1 kg of ice into water. This process takes place at a constant temperature of 0^0 C. This kind of heat required for this process is called the *Latent Heat of Fusion* (L_f). On the other hand, the quantity of heat required to completely change water into vapour at constant temperature of 100^0 C is called *Latent Heat of Vaporization* (L_v).

Mathematically, heat energy required to cause change of state is given as;

$Q = mL$

where L is the latent heat.

Example 5. Find the quantity of heat energy needed to melt 1.5 kg of ice completely to water. [take $L_f = 3.36 \times 10^5$ J Kg^{-1}]

Solution

$Q = mL = 1.5 \times 3.36 \times 10^5 = 5.04 \times 10^5 J$

Example 6. Find the quantity of heat energy required to melt 0.4 kg of ice at -5°C to water at 40°C.

Solution

In solving this question, three stages are involved.

1. Heat needed to raise the temperature of ice from -5°C to 0°C, $Q_1 = mc\Delta\theta$
2. Heat needed to melt ice at 0°C to water at 0°C, $Q_2 = mL_f$
3. Heat required to raise the temperature of water from 0°C to 40°C, $Q_3 = mc\Delta\theta$

$Q_1 = mc\Delta\theta = 0.4 \times 4186 \times (0 - (-5)) = 8372 J$

$Q_2 = mL_f = 0.4 \times 3.36 \times 10^5 = 1.344 \times 10^5 J$

$Q_3 = mc\Delta\theta = 0.4 \times 4186 \times (40 - 0) = 66976 J$

The total heat energy required to melt ice from -5°C to water at 40°C is;

$Q = Q_1 + Q_2 + Q_3 = 8372 + 1.344 \times 10^5 + 66976 = 209748 J$